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Research Article

An Eco-friendly Porous Nanocomposite Fabric-Based Triboelectric Nanogenerator for Efficient Energy Harvesting and Motion Sensing

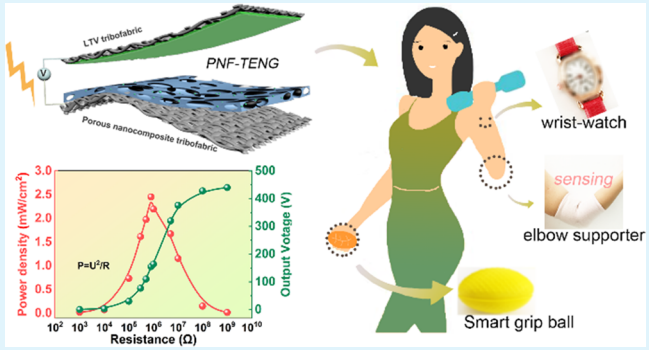
Zhiqing Bai, Yunlong Xu, Jiecong Li, Jingjing Zhu, Can Gao, Yao Zhang, Jing Wang,* and Jiansheng Guo*

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ABSTRACT: A wearable and effective tribopositive material, especially an economical and eco-friendly triboelectric fabric developed from biomaterials, is highly crucial for the development of green wearable triboelectric nanogenerators. In this work, we design a porous nanocomposite fabric (PNF) with strong charge accumulation capacity through a facile dry-casting method and use it as a tribopositive material to construct attractive wearable triboelectric nanogenerators (abbreviated as TENGs). Specifically, the porous nanocomposite is developed by the incorporation of nano-Al_2O_3 fillers into cellulose acetate networks. By adjusting the concentration of casting solution and the content of nano-Al_2O_3 fillers, we systematically engineer the physical properties of the PNF for obtaining a large triboelectric charge yield. When a 10 wt % solution concentration and 10 wt % nanofiller content are adopted for the PNF, the corresponding PNF-TENG can deliver an electrical performance of ~2.5 mW/cm^2 on a 0.8 MΩ external resistor. This remarkable output can be ascribed to the synergistic effect between the appropriate porous network and improved dielectric properties of the nanocomposite. Moreover, the PNF-TENG also exhibits good reliable electrical outputs under multiple stain-washing measurements or after experiencing cyclical contact–separation 13,500 times. Also, the device is capable of charging various capacitors, lighting LED arrays, and driving commercial wrist watches and is proven to be an efficient and reliable green wearable power source. Furthermore, a PNF-TENG-based elbow supporter and a grip ball, as self-powered sensors, are proposed to realize real-time detection for human actions during sports exercise. This work proposes an eco-friendly nanocomposite fabric as an effective tribopositive material, verifies the feasibility of developing environmentally friendly wearable power sources and sensors, and provides new insights into the design of green wearable triboelectric nanogenerators.

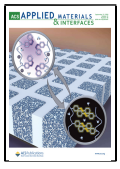


KEYWORDS: *triboelectric nanogenerator, cellulose acetate, nano-Al_2O_3 fillers, porous nanocomposite fabric, triboelectric charge yield, environmental friendliness*

INTRODUCTION

Nowadays, sustainable energy-harvesting technology has gained considerable attention not only to alleviate issues of global energy consumption and the ecological environment but also to continuously provide a power supply for micro-electronic devices and wireless sensors in emerging applications.[1−5] In 2012, triboelectric nanogenerators (TENGs), as one of the most promising mechanical energy harvesters, were proposed for the first time by Wang and co-workers and are capable of efficiently scavenging ubiquitous mechanical energy and converting it to electrical energy.[6] In comparison to other technologies, TENGs are advantageous in terms of high efficiency, large power output, light weight, easy fabrication, diverse material selection, etc. and thus have been extensively applied in power supply devices and self-powered sensing.[7−11] The working mechanism of TENGs is based on the triboelectric effect and electrostatic induction between two dissimilar materials with different electron affinities.[12] The triboelectric property of materials plays an essential role in the electrical performance of TENGs because it is related to the total yield of accumulated tribocharges in tribomaterials. Thus, the structure design and selection of tribomaterials are a top priority for developing high-performance TENGs. At present, a majority of demonstrated TENGs, especially for wearable devices, are widely fabricated by choosing various synthetic polymers or metallic materials, such as polytetrafluoroethylene (PTFE),[13] poly(vinylidene fluoride) (PVDF),[14] poly-

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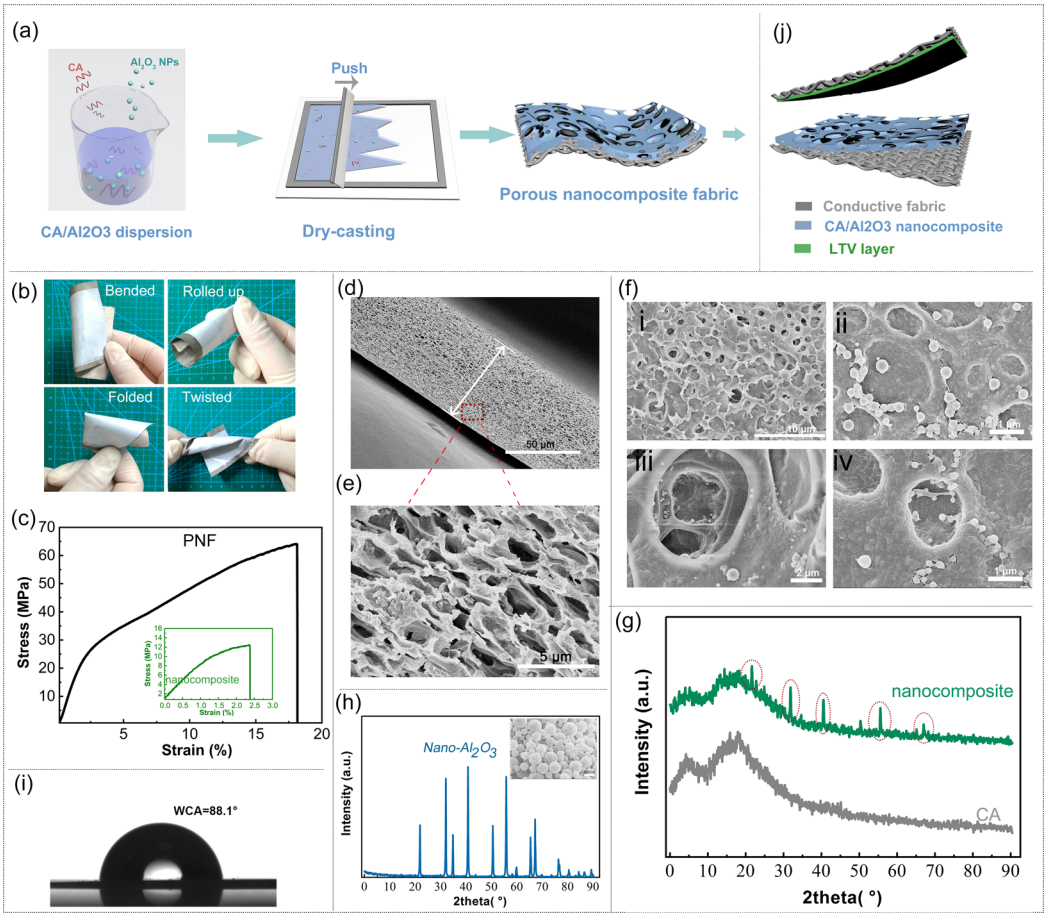


Figure 1. Preparation, characterization of the porous nanocomposite fabric, and structure of the PNF-TENG. (a) Schematic illustration of the fabrication for the PNF. (b) Digital images of the PNF, showing the softness and deformability. (c) Stress−strain curves of the PNF and porous nanocomposite. (d, e) Cross-sectional SEM images of the optimal porous nanocomposite. (f) SEM images of the surface morphology at different amplitudes and locations on the surface of the nanocomposite with 10 wt % nano-Al_2O_3 and 10 wt % CA concentration. (g, h) XRD spectra of Al_2O_3, CA, and the nanocomposite. The inset in (h) shows the rough surface of Al_2O_3 nanoparticles. (i) Water contact angle of the PNF. (j) Structural design of the PNF-TENG.

(dimethylsiloxane) (PDMS),[15,16] polyethylene terephthalate (PET),[17] polyamides (PA),[18] copper (Cu),[17] and aluminum (Al).[19] These commonly used dielectric materials are nonbiocompatible, nonrenewable, and nonbiodegradable, which not only limits their applications in wearable and biomedical applications but also causes severe electronic waste (e-waste) at the end of the service period. Furthermore, compared with numerous tribonegative materials, available tribopositive materials, especially nontoxic biodegradable and recyclable biomaterials, are greatly limited. Consequently, to explore sustainable tribopositive biomaterials for advancing green wearable TENGs is highly desirable given the human friendliness and environmental management.

Cellulose acetate (abbreviated as CA), as abundant biological resources, is considered an attractive tribomaterial due to its nontoxicity, non-irritation, good processability, film-forming ability, cost-effectiveness, biodegradability, and biocompatibility.[20−22] In general, CA biomaterial is spun into an electrospun nanofibrous membrane with the aid of electrospinning technology, which is used for the construction of green wearable TENG.[23] Although cellulose acetate nanofibrous membrane-based TENGs have great characteristics such as flexibility, high surface contact area, and light weight, they find it difficult to meet the mechanical durability need for human daily movement. Moreover, electrospinning technology usually involves complex procedures, having disadvantages of being inefficient and time-consuming and showing difficulties in mass production. Additionally, compared with a PA polymer, which is the most frequently used tribopositive materials, CA exhibits poor triboelectric properties and limited yield of tribocharge accumulation, which leads to the poor electrical performance of TENGs. How to realize a large tribocharge yield in cellulose acetate membranes via a simple preparation route for developing economically feasible and ecologically friendly TENGs with good performance is still a challenge.

In this regard, we reported a green tribopositive nanocomposite fabric based on a porous CA/Al_2O_3 membrane and a flexible conductive fabric and it was used to construct green wearable triboelectric nanogenerators. To improve the electrical performance of a porous nanocomposite fabric-based TENG (abbreviated as PNF-TENG), the concentration of CA solution and the nano-Al_2O_3 content in the PNF were discussed and optimized systematically. The optimal TENG device could deliver an open-circuit voltage (V_{OC}) of 448 V and an instantaneous power density of 2.5 mW/cm^2. Moreover, the washability, mechanical flexibility, and durability of the PNF-TENG were evaluated and it was found that it

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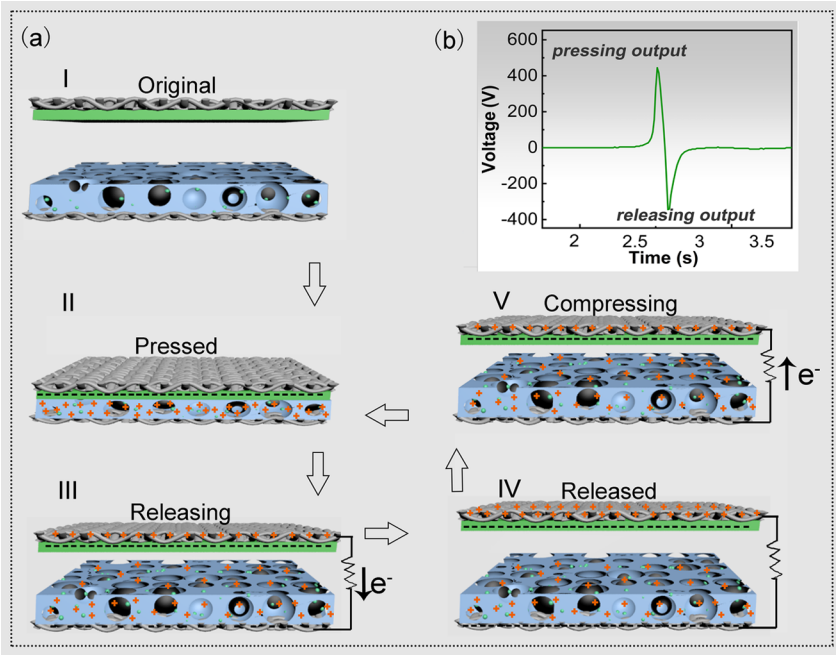


Figure 2. (a) Schematic illustration of the power generation principle of the PNF-TENG. (b) Typical output voltage signal for one pressing and releasing cycle.

exhibited reliable electrical performance and wearability. Finally, the great application potential of the PNF-TENG as an efficient green wearable power source and self-powered sensor was demonstrated through portable electronics powering and fitness exercise signal monitoring.

EXPERIMENTAL SECTION

Synthesis of a Porous Nanocomposite Fabric. CA powders were dissolved in acetone/water mixed solvents (85/15, v/v) to obtain 10 wt % transparent CA solutions. Different amounts (5, 10, 20, 40, 60 wt %) of nano-Al_2O_3 fillers were dispersed in acetone/water/dispersant solutions by alternate stirring and ultrasonic vibration. Subsequently, the finely dispersed nanoparticles were mixed with the CA transparent solutions and stirred vigorously at 60 °C for 60 min. When the excess acetone/water solvent was evaporated, the homogeneous dispersions were quickly coated onto a prepared groove model and the undried membranes were allowed to heat for obtaining the porous nanocomposite. Finally, the nanocomposites were adhered to a conductive fabric for the preparation of porous nanocomposite fabrics. Additionally, pure CA tribomaterials with different mass fractions (6, 8, 10, 14 wt %) were prepared in a similar way.

Construction of the PNF-Based Triboelectric Nanogenerator. According to the triboelectric series, low-temperature vulcanized (LTV) silicone was employed as the tribonegative material due to strong electronegativity.[24] First, the conductive fabric was ultrasonically cleaned in acetone, alcohol, and deionized (DI) water to remove impurities. Then, the LTV base monomers and cross-linkers were mixed in a mass ratio of 1:1 in a beaker. Subsequently, the uniform mixture was blade-coated onto the cleaned conductive fabric and was heated at 80 °C for obtaining the tribonegative material. Afterward, tribonegative materials and tribopositive materials were cut into squares with different dimensions and fixed onto a paperboard substrate face-to-face for completing the fabrication of PNF-TENGs.

Characterization and Measurement. Surface topographies were investigated using a field emission scanning electron microscope (Hitachi SU8010, Japan). Fourier transform infrared spectroscopy (FTIR) was used to evaluate the surface chemical composition of samples. The corresponding water contact angle (WCA) was determined using an optical contact angle measurement analysis system (OCA15EC, German). The mechanical properties were measured using an electronic tensile machine. The phase structures of CA, Al_2O_3, and the porous nanocomposite were determined by X-ray powder diffraction (XRD) using Cu Kα radiation (λ = 1.5406 Å). The electrical properties of the TENGs were recorded using a digital oscilloscope and a programmable digital electrometer (Keithley model 6514, America).

RESULTS AND DISCUSSION

Fabrication and Characterization of the PNF. The fabrication process of the PNF is schematically depicted in Figure 1a. In brief, fine nano-Al_2O_3 fillers were ultrasonically dispersed in acetone/water/dispersant solvent and then were uniformly mixed with CA solution for obtaining CA/Al_2O_3 dispersion. Afterward, the prepared nanodispersion was poured into a groove model and the dried sample was adhered to a conductive fabric with a plain weave structure (Figure S1 in the Supporting Information). Figure 1b presents a piece of as-made PNF. It can be found that the PNF could be subjected to various mechanical stimuli, such as bending, folding, rolling up, and twisting, exhibiting good deformability and flexibility. Further, the stress−strain measurement in Figure 1c indicated that the PNF has a good breaking extension and an ability to withstand greater external forces, much larger than that of the individual CA/Al_2O_3 nanocomposite. Moreover, the micromorphology of the PNF was examined, as shown in Figure 1d,e. It is observed that the cross-sectional morphology of the CA/Al_2O_3 nanocomposite exhibits a three-dimensional ordered porous structure with better orientation. Such a porous structure could be attributed to the microphase separation in a CA/Al_2O_3/acetone/water mixture.[25,26] Also, a magnified view in Figure 1e clearly shows that the pore skeleton is rough and is shaped like little waves. The surface morphology of the CA/Al_2O_3 nanocomposite was also examined, as shown in Figure 1f. In comparison to the pure CA membrane with a smooth pore skeleton (Figure S2 in the Supporting Information), the nanocomposite exhibits a

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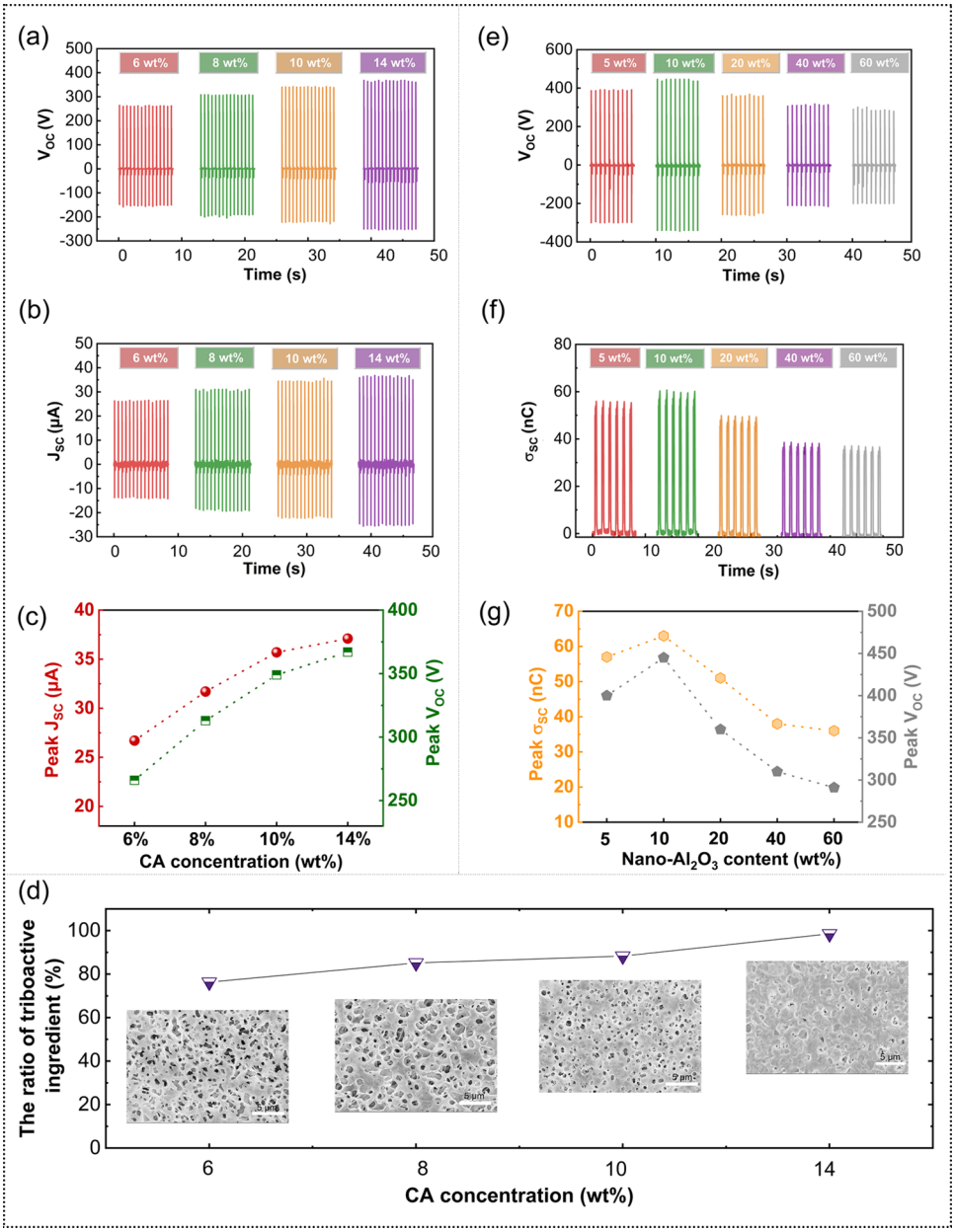


Figure 3. Effect of the concentration of CA solution and nano-Al_2O_3 content on the electrical outputs. (a) V_{OC} and(b) J_{SC} of the CA-TENG prepared by a series of porous CA membranes. (c) Curves showing the relationship between the electrical outputs and concentration of casting solution. (d) Surface morphology and triboactive ingredient ratio of pure CA membranes with different solution concentrations. (e) V_{OC} and (f) σ_{SC} of the PNF-TENG at different nano-Al_2O_3 fillers contents. (g) Dependence of V_{OC} and σ_{SC} of the PNF-TENG on nano-Al_2O_3 filler content.

rougher surface with a hierarchical morphology on which microscale pores or cavities with different depths and nano-Al_2O_3 particles are randomly distributed. Note that appropriate porous structures and dispersed nanoparticles favor the power enhancement of TENGs, as discussed later. To further reveal the structure and dispersion of nano-Al_2O_3 fillers into CA porous networks, the X-ray diffraction patterns of the porous nanocomposite, pure CA membrane, and nano-Al_2O_3 were examined, as shown in Figure 1g,h. It is observed that the XRD pattern of the porous nanocomposite exhibits the characteristic peaks of nano-Al_2O_3 fillers at 2θ = 25.6, 34.9, 43.1, 57.2, and 68.1°, which indicates that the nano-Al_2O_3 fillers were uniformly distributed in the cellulose acetate networks. The inset in Figure 1h is the SEM image of the surface morphology of nano-Al_2O_3, with a rough wrinkled surface and a diameter of about 100 to 700 nm. Additionally, the water contact angle (WCA) on the surface of the PNF and CA was measured and compared, as shown in Figure 1i and Figure S3 in the Supporting Information. The WCA of the PNF surface was enhanced up to ~88.1°, much larger than that of the pristine CA membrane (~71.3°). The increased hydrophobicity implies that the PNF is less sensitive to water molecules in the atmosphere, which would lead to a reduced tribocharges loss for the surface of the PNF and improve the environmental adaptability of the device.[27,28] Considering all of these merits, the PNF was considered as an attractive tribopositive material for developing green wearable TENGs.

Working Principle of the PNF-TENG. The structure design of the PNF-based TENG is demonstrated in Figure 1j in which flexible low-temperature vulcanized (LTV) silicone

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ACS Appl. Mater. Interfaces 2020, 12, 42880−42890

was selected as the tribonegative pair and the lightweight conductive fabric was chosen as an induction electrode for meeting the requirements of wearability. Similarly, the triboelectricity generation mechanism of the PNF-TENG is based on the coupling effect of contact electrification with electrostatic induction,[29,30] which is explained in Figure 2a. Before physical contact between the PNF and LTV tribomaterials, no charge transfer occurs (Figure 2aI). When two tribomaterials were brought together under the compression force, electrons will inject from the PNF to the LTV layer owing to the contact electrification effect (Figure 2aII), leading to opposite charges on the surfaces of the triboelectric materials. Subsequently, as two tribomaterials start to separate from each other, an electric potential difference between the PNF and LTV tribomaterials would be gradually established due to the electrostatic induction effect. At this moment, electrons flow from the top fabric electrode to the bottom fabric electrode until an electrostatic equilibrium was reached (Figure 2aIII,IV). Once the PNF-TENG is pressed again, the previous potential difference would disappear, which caused a backflow of free electrons in an inverse way (Figure 2aV). One typical electrical signal of the PNF-TENG in one pressing and releasing cycle is also presented in Figure 2b. It is worth noting that the charge generation and accumulation for the as-made PNF-based TENG are closely related to the appropriate porous structure and nanofillers in the nanocomposite fabric. On the one hand, the positive tribocharges were generated not only on the top surface of the PNF tribomaterial but also on the surface of the inner pores, which would contribute to forming a larger electrical potential difference once the device was released from pressing.[27,31] On the other hand, the distance between the top and bottom electrodes could be decreased due to the deformation of the inner pores of the PNF. More importantly, the dielectric constant of the PNF tribomaterial could be greatly enhanced owing to the addition of nano-Al_2O_3 fillers.[32,33] Consequently, the total capacitance for the PNF-TENG would be enlarged, which supported the enhancement of the electrical performance of the device.

Optimization of Electrical Performance. To achieve the high electrical performance of the PNF-TENG, the effects of the concentration of CA solution and nano-Al_2O_3 content on electrical outputs were systematically explored in this section. To start, a group of CA casting solutions with different concentrations of 6, 8, 10, and 14 wt% was prepared for obtaining porous CA membranes. Subsequently, these samples were paired with LTV tribonegative fabrics for constructing corresponding TENGs and these devices were fixed onto paperboards, attaching with a customized pressing machine for subsequent electrical performance measurements. As shown in Figure 3a,b, CA-TENGs delivered considerable output voltage and current signals. Particularly, the open-circuit voltage (V_{OC}) and short-circuit current (J_{SC}) of the device with the 14 wt % porous CA membrane could be up to 368 V and 37.1 μA, respectively. Compared with the PNF-TENG with the 6 wt% CA membrane, the V_{OC} and J_{SC} were enhanced by about 39 and 38%, respectively. The considerable output enhancement could be attributed to the reasonable adjustment of the pore content via an increase in the concentration of triboactive ingredients in porous membranes. Figure 3d, and Figures S4 and S5 in Supporting Information revealed the bulk ratio of CA in different porous CA membranes, pore size, and their distributions. With the CA concentration increased from 6 to 10 wt %, the bulk ratio of the CA triboactive ingredient (V_{CA}/V_{total}, where V_{CA} and V_{total} are the volume of the CA skeleton and the total volume of the porous nanocomposite, respectively) increased from ~76 to ~88%, which provides a larger surface contact surface and electron donors for contact electrification of the device, leading to more tribocharge accumulation. In this study, it is interesting to note that although the pore content decreased with increasing CA concentration, the electrical outputs were still greatly enhanced. This was because the pore content (V_{pore}/V_{total}, where V_{pore} is the volume of the pore or cavities in the porous nanocomposite) at low CA concentration may already be well above the optimal threshold, in which case excessive pores could not exert a positive effect on the electrical properties of TENGs but rather a negative effect[34] and the CA triboactive ingredient in the porous CA membrane is dominant in the improvement of electrical performance. For a visual presentation, the output peak V_{OC} and J_{SC} as a function of the concentration of CA solution are displayed in Figure 3c. The electrical outputs of CA-TENGs exhibit an approximately linear increase first with the CA concentration increased from 6 to 10 wt %. However, the CA concentration is further increased, and the corresponding TENGs exhibit relatively stable electrical output. This could be attributed to the balanced result between the volume ratios of CA triboactive ingredients and pore content in the porous CA membrane. Herein, considering the high viscosity for the 14 wt % CA solution, the 10 wt % one was selected for the subsequent experiments for performance optimization.

Further, porous nanocomposites with different contents of nano-Al_2O_3 fillers (5, 10, 20, 40, 60 wt %) were prepared for the construction of various PNF-TENGs. The measured V_{OC}, σ_{SC}, and J_{SC} of devices were recorded and compared in Figure 3e,f and Figure S6 in the Supporting Information. It is observed that although the prepared CA-TENG using the pure porous CA membrane has achieved considerable electrical outputs, and the positive effect of nano-Al_2O_3 doping on the electrical performance of TENGs is still notable. When the 5 wt % nano-Al_2O_3 fillers only were introduced into the porous nanocomposite fabric, the peak V_{OC}, J_{SC}, and short-circuit charge transfer (σ_{SC}) could leap up to 400 V, 40.3 μA, and 55 nC, respectively, exhibiting a remarkable enhancement. Nano-Al_2O_3, as biocompatible ceramic fillers, is commonly used to develop functional nanocomposites with a high dielectric constant. Herein, the dielectric constant of the PNF tribopositive material was improved resulting from the even distribution of nano-Al_2O_3 in cellulose acetate networks.[32,35] According to the formula of maximum transferred charge density (σ)[36,37] $\sigma = \frac{\sigma_0 D_{gap}}{D_{gap} + D_{film}/\varepsilon_{film}}$, where ε_{film} is the dielectric constant of the porous nanocomposite, σ_0 is the transferred charge density at the equilibrium, D_{gap} is the gap distance, and D_{film} is the thickness of the nanocomposite, the enhanced dielectric constant of the PNF tribomaterial would lead to the increased triboelectric charge density, thereby realizing an improvement in electrical outputs. When 10 wt % nano-Al_2O_3 fillers were introduced into the porous nanocomposite, the corresponding device achieved the highest electrical outputs with a V_{OC} of 448 V, J_{SC} of 45 μA, and σ_{SC} of 63 nC, which were about 1.3, 1.3, and 1.5 times higher than that of the pure CA membrane-based TENG, respectively. However, as the content of nano-Al_2O_3 fillers was further increased, the electrical output of the PNF-TENG exhibits a dramatic

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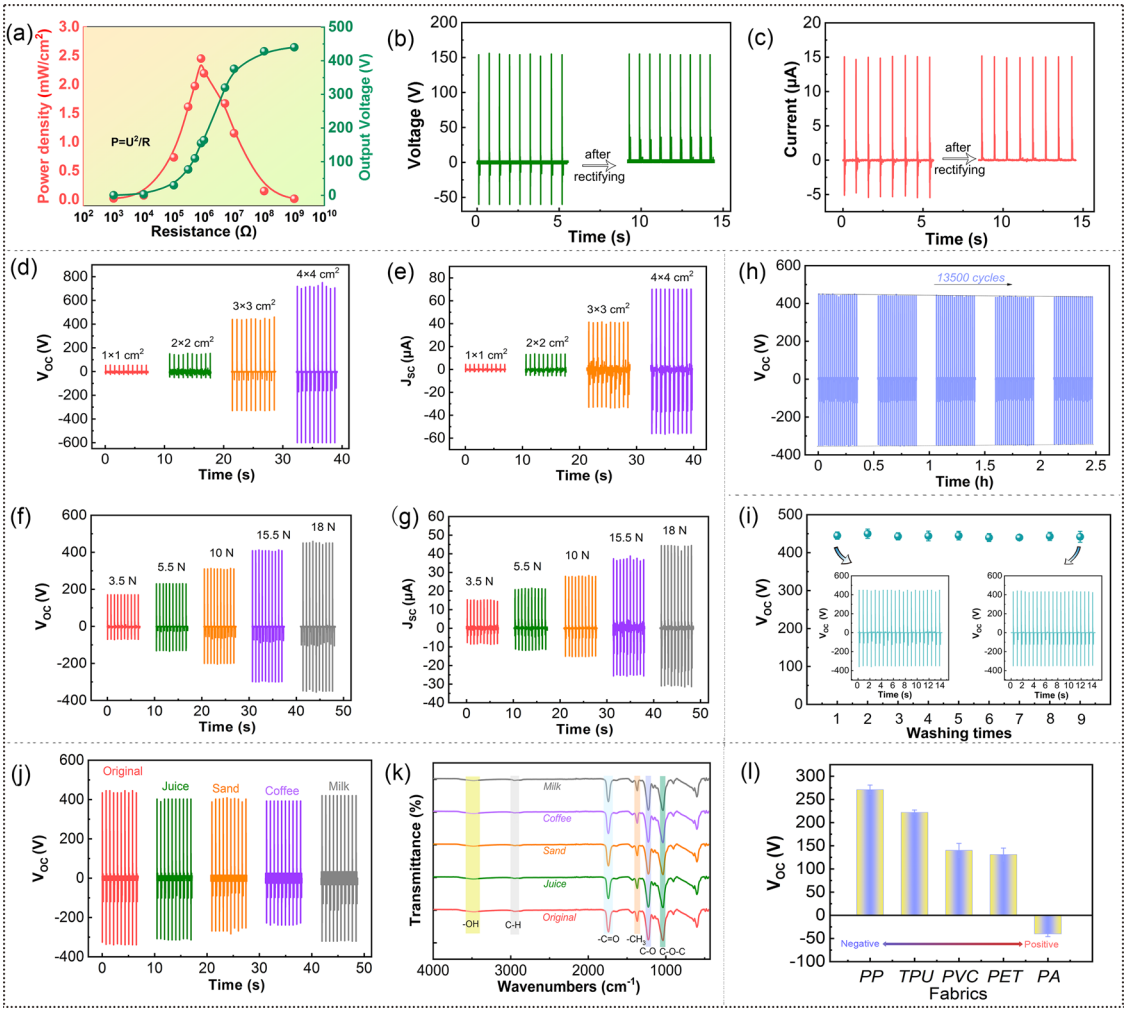


Figure 4. Evaluation of electrical outputs, durability, and washability for the PNF-TENG with 10 wt % nano-Al_2O_3 and 10 wt % CA concentration. (a) Curves of output voltage and instantaneous power density versus a series of external load resistances. (b) Output voltage and (c) output current of the device before and after rectification at an external resistance of 0.8 MΩ. Dependence of V_{OC} and J_{SC} on the different dimensions of the (d, e) PNF and (f, g) external impact forces. (h) Stability of electrical outputs being subjected to contact−separation movements for approximately 2.5 h. (i) Voltage output of the corresponding devices after repeated washing times. (j) Electrical output of the PNF-TENGs when the PNF is stained by common stains and then washed for 30 min. (k) FTIR analysis of the PNFs before and after being stain-washed. (l) Output signals of PNF-TENGs prepared by the tribopositive PNF paired with commonly used tribonegative fabrics.

Table 1. Comparison of the Eco-friendly Triboelectric Nanogenerators in Recent Years with PNF-TENGs

eco-friendly TENGs	force (N)	frequency (Hz)	voltage (V)	current (μA)	power (mW/m^2)	reference
cellulose II aerogel/PTFE	40	4	65	1.86	127	39
chitosan-diatom/FEP	8	5	150	1.02	15.7	40
starch/PTFE			1.2		170	41
BC-$BaTiO_3$/PDMS	42	2	181	21	4800	42
MCC-PVA/Kapton	4−5	5	600	50	84,500	43
CNF-PEI/PVDF nanofiber	6		106.2	9.2	13,300	44
gelatin/PLA	50	5	500	16.96	5000	45
CNF/AgNWs	16.8	1	21	2.5	693	46
PCL-GO/paper	13	3	120		72.5	47
CA-Al_2O_3/LTV	18	1.5	448	45	25,000	this work

decrease. This might be related to the lower effective contact area resulting from inadequate surface contact between LTV and the CA triboactive ingredient in the PNF, which is attributed to the excessive cover of nano-Al_2O_3 on the surface of the PNF and the increased hardness of the nanocomposite after doping of nano-Al_2O_3 inorganic ceramic fillers (Figure S7 in the Supporting Information). Meanwhile, the improvement in electrical outputs via introducing nano-Al_2O_3 fillers to increase the dielectric constant is no longer prominent. Furthermore, Figure 3g clearly and intuitively reveals the strong dependence of output V_{OC} and σ_{SC} on nano-Al_2O_3 fillers in the PNF and demonstrates that 10 wt % nano-Al_2O_3 is a critical point in electrical output enhancement of the PNF-TENG. According to the above discussion, the corresponding

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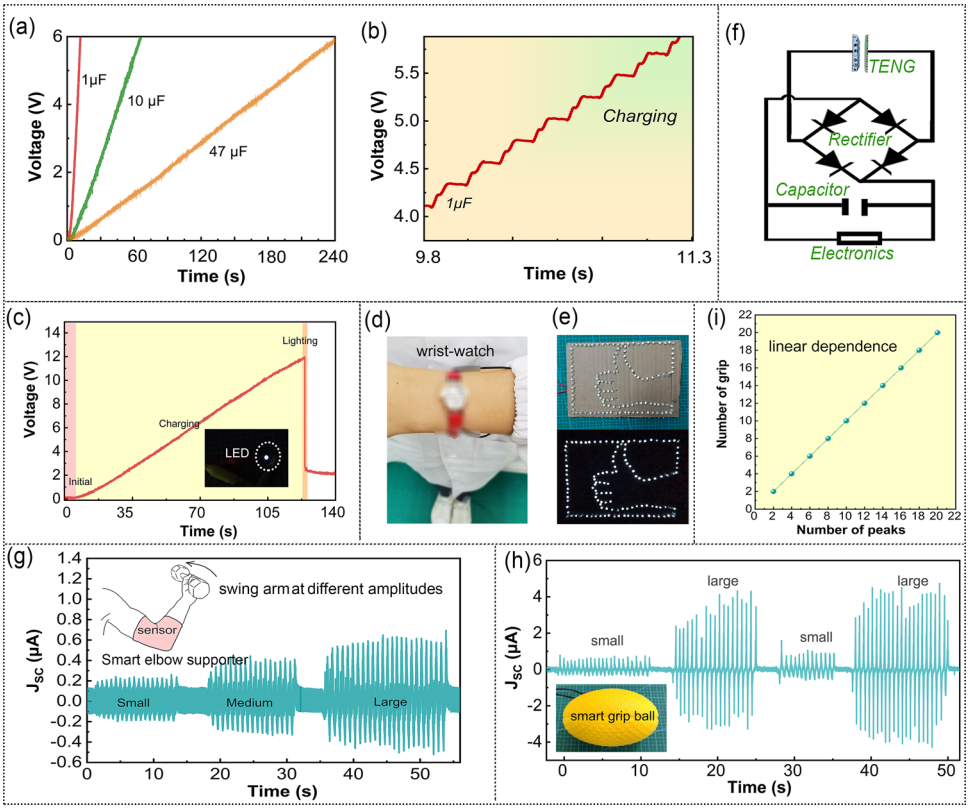


Figure 5. Demonstration of the PNF-TENG as a wearable power source and self-powered sensor. (a) Voltage charging curves of capacitors with different capacitances (1, 10, 47 μF). (b) Step voltage rise of the capacitor with a capacitance of 1 μF. (c) Charging and discharging curves of a capacitor (4.7 μF), with an inset showing a lit LED powered by the capacitor. Demonstration of (d) driving a wrist watch and (e) the LED arrays with a "thumb" sign powered by the PNF-TENG. (f) Simplified circuit diagram showing electronics charged by the PNF-TENG with the aid of a bridge rectifier. (g) Detected current output from the PNF-TENG-based sports elbow supporter. (h) Detected signals from a PNF-TENG-based homemade grip ball during human grip strength testing. (i) Linear relationship between the number of output peaks and the number of grips.

TENG sample could exhibit the optimal output performance when the 10 wt % nano-Al_2O_3 fillers were adopted for the preparation of the PNF.

Electrical Outputs and Washability. In view of the practical use of the PNF-TENG, the electrical output strongly depends on the resistance of the external load itself. Consequently, the output voltage and instantaneous power density measurements at varied external load resistances were carried out in detail. It is observed that as the load resistance increased from 0.001 to 1000 MΩ, the output voltage exhibited a gradual upward trend (Figure 4a), obeying Ohm's law. The power density was calculated according to the equation $P = U^2/RS$, where U is the output voltage under the corresponding loading resistance R and S is the contact area of tribopairs. The maximum instantaneous power density of the PNF-TENG could reach 2.5 mW/cm^2 at a resistance of 0.8 MΩ, as illustrated in Figure 4a. It is worth mentioning that such electrical output performance of the PNF-TENG is comparable to or superior to that of many previous eco-friendly TENG devices (Table 1). Figure 4b,c shows the output performance including an output voltage of ∼158 V and output current of ∼15 μA over the optimum impedance of 0.8 MΩ, from which we observed that the inherent pulsed alternating outputs obtained from the PNF-TENG are easily converted into stable direct outputs with the aid of a rectification circuit. Furthermore, we also investigated the electrical outputs at different sizes of tribomaterials and under different impact forces for better evaluation of the output performances of the PNF-TENG. As displayed in Figure 4d,e and Figure S8a in the Supporting Information, V_{OC} and J_{SC} exhibit an outstanding increase with the dimensions of the tribomaterial changing from 1 × 1 to 4 × 4 cm^2 owing to greater tribocharge production at the larger contact area. Moreover, the electrical outputs of the PNF-TENG also exhibit a strong dependence on the external forces, as shown in Figure 4f,g and Figure S8b in the Supporting Information. Under a stronger impact force, the device could output higher V_{OC} and J_{SC} due to a larger deformation between the PNF and LTV tribofabrics.[38] Furthermore, the output durability and stability of the PNF-TENG were evaluated under many contact-separation movements and long-term exposure to air, as shown in Figure 4h and Figure S9 in the Supporting Information. The V_{OC} profile of the PNF-TENG did not exhibit noticeable fluctuation during operation for 13,500 cycles. Also, the wearable device did not display measurable degradation in output voltage even after exposure to air for 1 month, compared with the pristine sample, confirming that the device has good reliability in long-term service.

Washability is an important performance index for wearable triboelectric nanogenerators in the real world. In this regard, the PNF was washed in a simulated laundering environment, in which DI water, common detergent, and a magnetic stir bar were mixed, as shown in Figure S10 (Supporting Information). The output performances of the PNF-TENG were measured after many repeated washing cycles in Figure 4i. It is observed that the washed PNF is in good condition and the

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ACS Appl. Mater. Interfaces 2020, 12, 42880−42890

corresponding V_{OC} signal of the device is well maintained, without measurable degradation, implying that the PNF is capable of withstanding long-term multiple cleaning cycles. To further evaluate their stain washability, five pieces of PNFs were successively stained by several common stains including juice, sandy soil, coffee, milk (Figure S11 in the Supporting Information), and they were cleaned in DI water. As shown in Figure 4j, the measured voltage of devices exhibited a minor decrease but was still maintained at above 87% of the output value of the original sample. The corresponding FTIR spectra of the pristine and stain-washed PNFs were examined in Figure 4k. It is observed that characteristic absorption peaks of PNF samples were similar, indicating no obvious change in surface chemical composition. This explains why the electrical output of the PNF-TENG remained relatively stable before and after being stain-washed. Note that those TENGs with the PNFs stained with milk and juice exhibit higher output signals than those of stained with sandy soil and hot coffee. The reason could be because the sandy soil and hot coffee in apertures between the conductive fabric and CA/Al_2O_3 nanocomposite are insoluble and difficult to be cleaned off relatively, leading to an inferior electrostatic induction effect of TENGs (Figure S12 in the Supporting Information). Additionally, the contact-electrification ability of the PNF in response to many daily-used fabrics including thermoplastic polyurethane (TPU), polyvinyl chloride (PVC) laminated fabrics, polypropylene (PP) non-woven fabrics, polyester (PET) fabrics was also examined. As shown in Figure 4l, compared with the PNF, PP, TPU, PVC, and PET easily withdraw electrons, with more tribonegativity; however, PA is more tribopositive. Note that the V_{OC} amplitude relies on the difference in the ability of electron-donating or electron-withdrawing groups when they paired with the PNF tribomaterial. Whatever the case is, these TENG devices exhibit considerable output voltage, which not only suggests the effectiveness of the PNF as a tribomaterial but also the wide selectivity of tribopairs for the PNF in practical applications.

Extensive Practicability of the Wearable PNF-TENG. In most cases, electronic equipment cannot be driven directly by the PNF-TENG due to intrinsic alternating output pulses. Many energy storage units such as a capacitor can be used to store triboelectric energy generated from the TENG device, providing a stable direct current (DC) power source for electronic devices. Therefore, it is very necessary to explore the charging behavior of the PNF-TENG for capacitors. Different commercial capacitors were charged by the PNF-TENG through a bridge rectifier, and the corresponding charging voltage−time curves are recorded in Figure 5a. It was found that capacitors with different capacitances of 1, 10, and 47 μF could be quickly charged to ∼6 V in approximately 12, 65, and 240 s, respectively, which indicates the rapid charging ability of the PNF-TENG. Figure 5b shows a step increase in the voltage of the capacitor due to every cycle of pressing and releasing on the PNF-TENG. Additionally, the charging−discharging curve of a capacitor with the capacitance of 4.7 μF was given concretely in Figure 5c. The charging voltage of the capacitor quickly increased to about 12 V in 122 s, at which point it was used to light up an LED to full brightness (Movie S1 in Supporting Information). With the PNF-TENG working again, the capacitor was charged for powering the LED again. These demonstrations suggest that the electrical energy scavenged by the PNF-TENG from human motion was not only sufficient to charge the capacitors but also to provide a powerful energy supply for microelectronics. Additionally, to further demonstrate the feasibility of the PNF-TENG as an effective power source, the device was attached to clothes for harvesting human motion energy and further to drive some commercial electronic devices. As shown in Figure 5d and Movie S2 in the Supporting Information, a commercial wrist watch can be driven by continuous running when the PNF-TENG with a size of 10 × 8 cm^2 was put in shoes. The simplified equivalent circuit is displayed in Figure 5f. Moreover, a self-made LED indicator with a "thumb" sign was easy to be lit up to full brightness by periodically holding the PNF-TENG (Figure 5e and Movie S3 in the Supporting Information).

Further, to explore the potential of the PNF-TENG as a wearable self-powered sensor, it was used to provide real-time and auxiliary monitoring of human movements in sports and fitness training. Herein, a kind of PNF-TENG-based smart sports elbow supporter was designed to detect the swing amplitude of an arm during a dumbbell exercise. As shown in Figure 5g and Movie S4 in the Supporting Information, the magnitude of current signals from the sports elbow supporter at a large swing amplitude is different from that at a small swing amplitude. When the swing amplitude of an arm, in turn, changed to small, medium, or large states, the value of detected output signals was enhanced gradually. The elbow supporter delivered a current of about 0.7 μA under a large swing amplitude, higher than the output at a small swing amplitude, which is because the elbow supporter could exert larger deformation under a large swing amplitude. Thus, the exercise intensity of body in dumbbell training process can be determined and assessed by the amplitude of the signal detected by the smart elbow supporter, which provides positive guidance for humans. for humans. Furthermore, we also designed a PNF-TENG-based smart grip ball for monitoring grip strength in real time. As displayed in Figure 5h and Movie S5, the electrical output amplitude of the grip ball exhibits a strong dependence on the grip strength. The distinct amplitudes of current signals would be generated when the grip ball experiences grips with different intensities. Also, the current signals exhibit good reproducibility under a certain grip strength, which affirms the excellent working reliability and stability of the PNF-TENG-based grip ball. Based on this, we could identify the intensity of grips according to the amplitude of the signal, which is beneficial to assessing body exercise conditions. The grip ball not only monitors the intensity of grips but also tracks and records the number of grips during the grip training process, as shown in Figure S13 and Movie S6 in the Supporting Information. It was observed that the number of grips is linearly related to the number of current signals in Figure 5i. When various numbers of grips (2, 4, 6, 8, 10, 12, 14, 16, 18, 20) were applied to the grip ball in succession, the corresponding numbers of electrical signals would be detected. These application demonstrations of a smart sports elbow supporter and a grip ball in sports and health exercises imply that the PNF-TENG has a bright future in the field of human motion sensing.

CONCLUSIONS

In summary, we have proposed a porous nanocomposite fabric by incorporating nano-Al_2O_3 fillers into cellulose acetate porous networks to develop wearable and green TENGs. This porous nanocomposite fabric was fabricated using a dry-casting method, and it was considered an effective tribopositive material. When a 10 wt % solution concentration and 10 wt %

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ACS Appl. Mater. Interfaces 2020, 12, 42880−42890

nano-Al_2O_3 fillers were incorporated in the PNF, the PNF-TENG could deliver the maximum electrical performance, with a V_{OC} of 448 V and power density of 2.5 mW/cm^2. The considerable outputs were attributed to the synergistic effect between the proper pore content, triboactive ingredient, and uniformly distributed nano-Al_2O_3 nanoparticles. Furthermore, the device also exhibited good mechanical strength, deformability, flexibility, and washability, which provides a guarantee for real-world applications. Based on these advantages, the PNF-TENG was effectively capable of charging commercial capacitors, powering wrist watches, and LED arrays and exhibited great potential as a wearable power source. Also, a PNF-TENG-based smart elbow supporter and a grip ball were designed and further employed for monitoring human motion states in real time during sports and health training, demonstrating the feasibility of the device as a promising self-powered sensor.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.0c12709.

SEM image of the conductive fabric as a flexible electrode, micromorphology of the pristine CA membrane, water contact angle of the pure CA membrane, pore diameter and the distributions under different CA concentrations, mean pore size of samples under different CA concentrations, J_{SC} of the device at different contents of nano-Al_2O_3 fillers, negative effect of nano-Al_2O_3 on electrical performance, dependence of electrical performance on the size of the sample and external impact forces, electrical outputs of the PNF-TENG before and after exposure to air for 1 month, PNF washed in a water/detergent mixture, PNFs with different stains, aperture states of the PNF before and after being stained, and output current of the grip ball at various numbers of grips (PDF)

Movie S1: driving a commercial LED to full brightness by the PNF-TENG (MP4)

Movie S2: powering a wrist watch by continuous running (MP4)

Movie S3: lighting a "thumb" LED array by the PNF-TENG (MP4)

Movie S4: output signals from the PNF-TENG-based sports elbow supporter (MP4)

Movie S5: detected electrical output for the grip strength using a smart grip ball (MP4)

Movie S6: corresponding signals of a grip ball under different numbers of grips (MP4)

■ AUTHOR INFORMATION

Corresponding Authors

Jing Wang − *College of Textile and Clothing Engineering, Dezhou University, Dezhou 253023, China*; Email: spu-1@163.com

Jiansheng Guo − *Key Laboratory of Textile Science and Technology, Ministry of Education, College of Textiles, Donghua University, Shanghai 201620, China*; orcid.org/0000-0002-8905-7040; Email: jsguo@dhu.edu.cn

Authors

Zhiqing Bai − *Key Laboratory of Textile Science and Technology, Ministry of Education, College of Textiles, Donghua University, Shanghai 201620, China*

Yunlong Xu − *Key Laboratory of Textile Science and Technology, Ministry of Education, College of Textiles, Donghua University, Shanghai 201620, China*

Jiecong Li − *Key Laboratory of Textile Science and Technology, Ministry of Education, College of Textiles, Donghua University, Shanghai 201620, China*

Jingjing Zhu − *Key Laboratory of Textile Science and Technology, Ministry of Education, College of Textiles, Donghua University, Shanghai 201620, China*

Can Gao − *Key Laboratory of Textile Science and Technology, Ministry of Education, College of Textiles, Donghua University, Shanghai 201620, China*

Yao Zhang − *Key Laboratory of Textile Science and Technology, Ministry of Education, College of Textiles, Donghua University, Shanghai 201620, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsami.0c12709

Notes

The authors declare no competing financial interest.

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